```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract StudentData {
    // Structure to represent a student
    struct Student {
        string name;
        uint256 age;
    }

    // Array to store multiple students
    Student[] public students;

    // Event to log student data addition
    event StudentAdded(string name, uint256 age);

    // Counter for the number of times Ether is sent
    uint256 public etherSentCount;

    // Fallback function to receive Ether and increment the counter
    receive() external payable {
        etherSentCount++;
    }

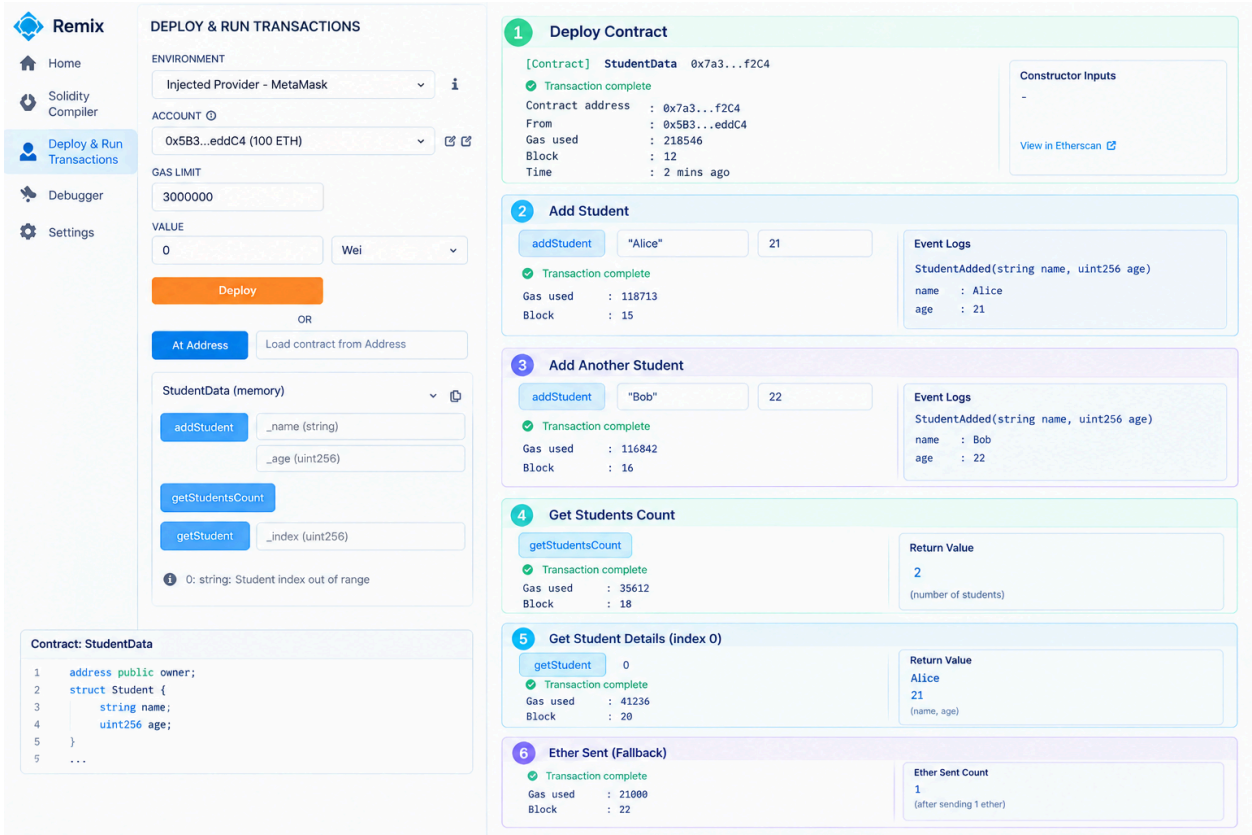
    // Function to add a new student
    function addStudent(string memory name, uint256 age) public {
        Student memory newStudent = Student(name, age);
        students.push(newStudent);
        emit StudentAdded(name, age);
    }

    // Function to get the count of students
    function getStudentsCount() public view returns (uint256) {
        return students.length;
    }

    // Function to get a specific student's details by index
    function getStudent(uint256 index) public view returns (string memory, uint256) {
        require(index < students.length, "Student index out of range");
        Student memory student = students[index];
        return (student.name, student.age);
    }
```

}

Output


Remix
Home
Solidity Compiler
Deploy & Run Transactions
Debugger
Settings
DEPLOY & RUN TRANSACTIONS
ENVIRONMENT
Injected Provider - MetaMask
ACCOUNT
0x5B3...eddC4 (100 ETH)
GAS LIMIT
3000000
VALUE
0
Wei
Deploy
OR
At Address
Load contract from Address
StudentData (memory)
addStudent
_name (string)
_age (uint256)
getStudentsCount
getStudent
_index (uint256)
0: string: Student index out of range
Contract: StudentData
address public owner;
struct Student {
string name;
uint256 age;
}
...
1 Deploy Contract
[Contract] StudentData 0x7a3...f2C4
Transaction complete
Contract address : 0x7a3...f2C4
From : 0x5B3...eddC4
Gas used : 218546
Block : 12
Time : 2 mins ago
Constructor Inputs
-
View in Etherscan
2 Add Student
addStudent
"Alice"
21
Transaction complete
Gas used : 118713
Block : 15
Event Logs
StudentAdded(string name, uint256 age)
name : Alice
age : 21
3 Add Another Student
addStudent
"Bob"
22
Transaction complete
Gas used : 116842
Block : 16
Event Logs
StudentAdded(string name, uint256 age)
name : Bob
age : 22
4 Get Students Count
getStudentsCount
Transaction complete
Gas used : 35612
Block : 18
Return Value
2
(number of students)
5 Get Student Details (index 0)
getStudent
0
Transaction complete
Gas used : 41236
Block : 20
Return Value
Alice
21
(name, age)
6 Ether Sent (Fallback)
Transaction complete
Gas used : 21000
Block : 22
Ether Sent Count
1
(after sending 1 ether)